THE DIAGONAL OF THE STACK OF CURVES IS REPRESENTABLE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Equation quot-equation-curves-over-proper-spaces (canonical tag 0D50). cited at proof line 4846. The fully faithful embedding.
- Lemma quot-lemma-spaces-diagonal (canonical tag 0D1F). cited at proof line 4848. Collected card: diagonal of the stack of spaces.

Source reference. Original equation 0D50: sources/stacks-chow/tags.txt, source lines 16470–16470. Cited at statement line 4846: the fully faithful embedding.

Source reference. Original lemma 0D1F: sources/stacks-chow/tags.txt, source lines 16345–16345. Cited at statement line 4848: corresponding diagonal for the stack of spaces.

Lemma 0.2. *The diagonal*

$$\Delta : \mathcal{C}urves \longrightarrow \mathcal{C}urves \times \mathcal{C}urves$$

is representable by algebraic spaces.

Proof. This is immediate from the fully faithful embedding (0D50) and the corresponding fact for $\mathcal{S}paces'_{fp,flat,proper}$ (Lemma 0D1F). □

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